

UNIVERSITY OF SZEGED
FACULTY OF AGRICULTURE

Engineering Internship Report

Plant Production and Horticulture Practice
Greenhouse Tomato Production

Student: *[Full Name]*

Course: Agricultural Engineering BSc

Internship place: *[Company name, greenhouse facility,
town/address]*

Internship period: *[start date – end date (6-week practice
period)]*

Internship leader: *[Name and professional status/title]*

Hódmezővásárhely
[Year]

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1 Circumstances of the Establishment of the Company

[Describe when and how the greenhouse company was founded, its legal form (e.g. private enterprise, family farm, cooperative), ownership history, and the motivation behind starting tomato production under glass/plastic.]

2 Organizational Structure of the Company

[Present the organizational chart (organogram) of the company: management, production supervisors, greenhouse staff, logistics/sales. Insert the organogram as a figure in the Appendix (uncomment the example in appendix.tex) and reference it here, e.g. Figure ??, once that figure exists.]

2.1 Research Institute Activity

[Not applicable – the internship took place at a commercial greenhouse operation, not a research institute or experimental station.]

3 Human Resource Characteristics of the Company

[Describe the number of employees, qualifications, seasonal vs. permanent staff, age structure, and training practices.]

4 Founding Capital, Fixed and Current Assets

[Give the booking value of founding capital, fixed assets (greenhouse structures, irrigation systems, vehicles) and current assets (inventories, receivables), as far as the company was willing to disclose.]

5 Liabilities and Financing

[Describe past loans/liabilities taken on by the company, their purpose (e.g. greenhouse construction, equipment purchase), and future financing needs and possibilities.]

6 General Description of the Plant Production Unit

[Describe the greenhouse farm as a whole: total covered area, number and type of greenhouse units, location, and how the tomato-growing unit fits into the company's broader operation.]

7 Ecological Conditions of Production

- **Climatic conditions:** *[regional climate, temperature/light regime, heating and ventilation needs]*
- **Production area:** *[total area under tomato cultivation, in hectares/m²]*
- **Soil properties:** *[soil analysis data if grown in soil]*
- **Growing media properties:** *[properties of substrate used, e.g. rockwool, coco coir, perlite, if soilless culture]*

8 Analysis of Production Activity and Products

[Introduce the tomato varieties/products grown (fresh market, processing, cherry/cocktail types, etc.), volumes produced, and how production activity is analyzed by the company (yield records, quality grading).]

9 Greenhouse Types

[Describe the greenhouse structures used: covering material (glass, polycarbonate, polyethylene film), structural type (Venlo, tunnel, etc.), and growing season(s) covered.]

10 Description of Production Profile

[Describe the production profile of the operation: tomatoes as the main crop, any secondary crops, and how production is scheduled across the year.]

11 Cultivated Varieties and Biological Bases

[Give a detailed description of the tomato varieties/hybrids cultivated, their biological characteristics (growth habit, ripening time, disease resistance), and seed/seedling sourcing.]

12 Cultivation Technology

12.1 Soil/Substrate Cultivation

[Describe soil preparation or substrate setup prior to planting.]

12.2 Greenhouse Preparation

[Describe preparation of the greenhouse for the crop cycle: irrigation system setup, plant support/trellising system, soil or substrate covering.]

12.3 Nutrient Supply

[Describe fertilization program: type of fertilizer/fertigation, dosage, and timing throughout the crop cycle.]

12.4 Planting Parameters

[Give planting time, row distance, plant spacing, and (if sown directly) sowing depth and seed rate.]

12.5 Plant Care

[Describe ongoing plant care: pruning, training, de-leafing, irrigation scheduling, pollination (e.g. bumblebee hives).]

12.6 Plant Protection

[Describe weed control and protection against diseases and pests (integrated pest management, biological control agents, chemical treatments used).]

12.7 Harvesting and Storage

[Describe harvesting method and frequency, grading, packaging, and post-harvest storage/cold chain.]

12.8 Greenhouse Culture Cultivation Technology Summary

[Summarize the overall cultivation technology cycle from planting to end-of-season crop removal.]

12.9 Seed Production, Processing and Variety Maintenance

[Not applicable – the internship focused on tomato production for the fresh/processing market, not sowing seed production.]

13 Machinery Park

[List the machinery used: tractors, implements, irrigation and fertigation equipment, climate control systems (heating, ventilation, CO₂ dosing, screens).]

14 Introduction of Experiments

[Describe any trials or experiments observed or assisted with during the internship (e.g. variety trials, new substrate or fertigation tests), if applicable.]

15 Work Management

15.1 Organizational Structure of the Farm

[Describe how the greenhouse farm is organized day-to-day.]

15.2 Personnel Conditions

[Describe manpower, education level of workers, and work fields/roles.]

15.3 Organizing Working Time

[Describe working hours, rest days and shift organization, especially during peak harvest.]

15.4 Work Organization Solutions

[Describe any specific work organization solutions used in the greenhouse (e.g. harvest crews, task rotation, quality checkpoints).]

16 Economic Analysis

[Present yields, revenue, costs, unit cost per kg of tomatoes, profitability, efficiency indices, and a simplified cash-flow plan. Present figures in a table (see Appendix).]

17 Marketing Directions of the Products

[Describe the marketing channels for the tomatoes (wholesale markets, supermarket chains, direct sale, export), opportunities, and conditions.]

18 Development Goals, Objectives and Plans

[Describe the company's development goals and plans, current management problems, and possible resolutions.]

19 Use of Subsidy

[Describe any subsidies/grants used by the company (e.g. EU agricultural subsidies) and how they are administered.]

20 Extension Services and Advocacy

[Describe extension services, advisory bodies or advocacy groups the company uses, and their impact on farming profitability.]

21 Developmental Aims and Plans

[Summarize the company's longer-term developmental aims and plans, and your own conclusions from the internship experience.]

Appendix A – Charts, Diagrams and Pictures

*[Insert the organogram, greenhouse layout, production/economic charts, and photographs here.
Example figure below – replace with your own image file.]*

Appendix B – Internship Leader’s Declaration

The Internship Leader’s declaration about the completion of the Engineering Internship

The undersigned *[name of internship leader]* certify that *[student name]*, student of Faculty of Agriculture, University of Szeged, spent her/his Engineering Internship from *[start date]* until *[end date]* at *[name and headquarters of the company/firm]*.

I confirm that he/she accomplished their professional duties, wrote a daily diary about the tasks which corresponds to reality.

I grade his/her professional activity to be: *[excellent (5) / good (4) / satisfactory (3) / pass (2) / fail (1)]*

My professional opinion about the student’s activities:

[...]

Date: *[...]*

[signature, stamp]